Chapter 24 INT 14H, Serial port handler

Use

Serial port control and I/O.

Entry

AH selects the function required, while other registers are used for parameters, as follows.

DX indicates the serial port. As **Hunter 16** is equipped with two, DX may be 00H (COM1) or 01H (COM2).

Exit

The AX register indicates the communications status as follows. All other registers are preserved.

Bits in AH indicates the line status as follows, when set:

Bit	Significance if set
7	Timeout (ignore other bits)
6	Transmit shift register empty
5	Transmit hold register empty
4	Line break detected
3	Framing error
2	Parity error
1	Overrun error
0	Data ready

Bits of AL indicate the modem handshake status as follows, when set:

Bit	Significance
7	Receive line signal detect
6	Ring Indicator (RI)
5	Data Set Ready (DSR)
4	Clear To Send (CTS)
3	Delta Receive line signal detect
2	Trailing edge RI (Ring Indicator)
1	Delta DSR
0	Delta CTS

Comments

IBM PC BIOS provides four functions:

00H Initialise port
01H Send a character
02H Receive a character
03H Get port status
04H Extended initialise port (PS/2-compatible)

Hunter 16 BIOS provides identical functions, plus a number of extensions, providing access to additional facilities such as higher speeds and protocols:

1EH Read communications parameters
1FH No effect
20H Select communications parameters
21H Extended control
22H Extended status
23H Control handshaking
24H Control CTS
25H Control line drivers
26H Control RI power up
27H Get installed protocols count
28H Get protocol name
29H Get protocol handle
2AH Extended protocol menu
2BH Get extended protocol parameters
2CH Control DTR

Initialise port

Entry

AH=00H
DX=port
AL=parameters

Exit

AX=status

Description

Configures the port specified by DX, according to the bits sets in AL:

D7	D6	D5	Baud
0	0	0	110
0	0	1	150
0	1	0	300
0	1	1	600
1	0	0	1200
1	0	1	2400
1	1	0	4800
1	1	1	9600

D4	D3	Parity
0	0	None
0	1	Odd
1	0	None
1	1	Even

D2	Stop bits
0	One
1	Two

D1	D0	Data bits
0	0	7
0	1	7
1	0	7
1	1	8

Send character

Entry
AH=01H
AL=character
DX=port

Exit
AX=status

Description
Write the character to the port specified by DX.

On return, if bit 7 of AH is 0, the character was sent, otherwise AX indicates the reason for non-transmission. (To check if the reason was a timeout, you must use the Check status' function.)

Input character

Entry
AH=02H
DX=port

Exit
AH=status
AL=character

Description
Reads a character from the port specified by DX.

If AH=0, the character has been read correctly and is in AL. Otherwise, AH indicates the reason for failure.

Check status

Entry
AH=03H
DX=port

Exit
AH=line status
AL=modem status

Description
Reads the RS232 line and modem status into AX.

Extended initialise port (+)

Entry

AH=04H
DX=port
BH=parity
BL=stop bits
CH=data bits
CL=baud rate

Exit

AX=status

Description

Configures the port specified by DX, according to the other registers:

CL	Baud
0	110
1	150
2	300
3	600
4	1200
5	2400
6	4800
7	9600
8	19200
9	38400

BH	Parity
0	None
1	Odd
2	Even
3	Odd
4	Even

BL	Stop bits
0	One
1	Two

CH	Data Bits
0	7
1	7
2	7
3	8

NOTE: This function is included for compatibility with the IBM PC-AT and IBM PS/2.

Read comms parameters (+)

Entry

AH=1EH
DS=segment address of buffer
BX=offset address of buffer
CX=channel number (00H=COM1, 01H=COM2)

Exit

Returns nothing

Description

Reads the configuration of the port specified by CX into the buffer specified by DS:BX. This information takes the following format:

Byte0	Effect
0	IBM communications
Non 0	Husky communications

Byte1	Port number
0	COM1
1	COM2

Byte2	Baud
0	110
1	150
2	300
3	600
4	1200
5	2400
6	4800
7	9600
8	19200
9	38400

Byte3	Data Bits
1	7
2	8

Byte4	Parity
0	None
1	Odd
2	Even

Byte5	Stop bits
0	One
1	Two

Byte6	Handshake
D7	ignored
D6=1	DCD enable
DS=1	DSR enable
D4=1	CTS enable
D3	ignored
D2=1	DTR enable
D1,D0=00	RTS off
D1,D0=01	RTS hold
D1,D0=10	RTS true
D1,D0=11	ignored

Byte7	Protocol
0	None
1	XON/XOFF
2	HWK3780 (option)

Byte8	Nulls
0..14H	0..20 Nulls after CR

Byte7	LF
0	Off
1	On

Byte10	Serig
0..7FH	Ignore this character
80H	Off

Byte11	Echo
0	Ignore this character
1	Off

Byte12	Transmit timeout
0	Off
1..3CH	1..60 seconds

Byte13	Receive timeout
0	Off
1..3CH	1..60 seconds

For details of Husky communications, see Part 2.

Select comms parameters (+)

Entry

AH=20H
DS=segment address of buffer
BX=offset address of buffer
CX=channel number (00H=COM1, 01H=COM2)

Exit

AX=0 - ok
AX=1 - invalid parameter(s)

Description

Configures the port specified by CX and byte 1 of the buffer, according to the contents of the buffer specified by DS:BX.
Note that both CX and byte 1 must specify the same buffer.

This buffer takes the following format:

Byte0	Effect
0	Select IBM communications
Non 0	Select Husky communications

Byte1	Port number
0	COM1
1	COM2

Byte2	Baud rate
0	110
1	150
2	300
3	600
4	1200
5	2400
6	4800
7	9600
8	19200
9	38400

Byte3	Data Bits
1	7
2	8

Byte4	Parity
0	None
1	Odd
2	Even

Byte5	Stop bits
0	One
1	Two

Byte6	Handshake
D7	ignored
D6=1	DCD enable
DS=1	DSR enable
D4=1	CTS enable
D3	ignored
D2=1	DTR enable
D1,D0=00	RTS off
D1,D0=01	RTS hold
D1,D0=10	RTS true
D1,D0=11	ignored

Byte7	Protocol
0	None
1	XON/XOFF
2	HWK3780 (option)

Byte8	Nulls
0..14H	0..20 Nulls after CR

Byte7	LF
0	Off
1	On

Byte10	Serig
0..7FH	Ignore this character
80H	Off

Byte11	Echo
0	Ignore this character
1	Off

Byte12	Transmit timeout
0	Off
1..3CH	1..60 seconds

Byte13	Receive timeout
0	Off
1..3CH	1..60 seconds

Byte14	RESERVED

Byte15	RESERVED

Byte16	RESERVED

Byte17	RESERVED

Byte18	**RESERVED**

NOTES: When used to configure COM2, inappropriate parameters are simply ignored.

For details of Husky communications, see Part 2.

Extended control (+)

Entry
AH=21H
AL=command
DX=port

Exit
AH-extended status (see 'Extended status, below)

Description
Used to control buffering. The commands are selected as follows:

AL	Command
AL=1	Force transmission of buffer
AL=2	Clear transmit buffer
AL=3	Clear receive buffer

Extended status (+)

Entry
AH=22H
DX=port

Exit
AH=extended status
BX=number of characters in input buffer
CX=number of characters in output buffer

Description
Used to obtain detailed error information from the current protocol, including the status of the transmit and receive buffers.

The standard protocols (NONE and XON/XOFF) return the most recent error code in AH. For details of the status information returned by other protocols, refer to the protocol documentation.

Protocols will also return error information in the RS232 status byte returned by the standard functions, using the bits which most closely

Control handshake lines (+)

Entry
AH=23H
BH=line
BL=level

Exit
Returns nothing

Description
Can be used to control the DTR and RTS handshake output lines directly. The specified level remains latched until changed by use of this function or normal communications handshaking.

BH	BL	Effect
0	0	set RTS low
0	1	set RTS high
1	0	set DTR low
1	1	set DTR high

Control CTS (+)

Entry
AH=24H
AL=control

Exit
Returns nothing

Description
Can be used to control sensitivity to the CTS handshake input.

AL	CTS
0	Disable
1	Enable

Control RS232 drivers (+)

Entry
AH=25H
AL=control

Exit
Returns nothing

Description

Can be used to turn on or turn off the RS232 line drivers, which are required for communication.

AL	Effect
0	Turn off
1	Turn on

The line drivers are normally left turned off when **Hunter 16** powers up, but turn on automatically when an application transmits, or **Hunter 16** receives handshaking inputs. They then remain turned on until **Hunter 16** turns off or you use this function.

Use this function to save power by turning off the drivers when communications has finished.

You can also use this function to turn on the drivers before connecting up to a remote system, as Hunter 16 may send one or two spurious characters while this is happening, which might confuse the remote system

Control RI power up (+)

Entry

AH=26H
AL=control

Exit

Returns nothing

Description

Receipt of the RI handshake will normally power up Hunter 16, allowing a suitable program to respond to devices such as autoanswer modems.

Use this function to enable or disable this mechanism:

AL	Effect
0	Enable power up
Non-0	Disable power up

This has no effect on reporting of the RI event (see Chapter 41).

Get installed protocols count (+)

Entry	AH=27H AL=count of extended protocols installed
Exit	AL=previous value of protocol count + number of resident installed protocols
Description	The purpose of this function is to determine how many protocols are currently installed in **Hunter 16**. It is used by the UTIL Comms menu option to determine how many protocols can be chosen from.

Get protocol name (+)

Entry	AH=28H AL=protocol handle DS:BX-pointer to 8 character buffer
Exit	AH=status (0=pass, FFH=fail)
Description	Use this function to read the name of the protocol with the handle stored in AL. On return, if AH is 0 the name is in the 8 character buffer pointed to by DS BX

Get protocol handle (+)

Entry	AH=29H DS:BX=pointer lo buffer containing name
Exit	AH=status (0=pass, FFH=fail) AL=handle

Description Use this function to read the handle for the protocol whose name is stored in the buffer pointed to DS:BX. On return, the handle is in AL if AH is 0.

Extended protocol menu (+)

Entry
AH=2AH
AL=protocol handle

Exit
AH=status (0=pass, FFH=fail)
AL=handle

Description Use this function to configure the protocol specified by the handle in AL, by displaying a menu.

Get extended protocol parameters (+)

Entry
AH=2BH
AL=protocol handle
DS:BX-pointer to 8 character buffer

Exit
AH=status (0=pass, FFH=fail)

Description Use this function to read parameters for the protocol with the handle stored in AL. On return, if AH is 0 the parameters are in the buffer pointed to by DS:BX.

Get/set DTR (+)

Entry
AH=2CH
AL=0 (get) or non-0 (set)
BL=0 (normal) or non-0 (on), if set

Exit
AH=0
BL=0 (normal) or non-0 (on), if get

Description DTR normally goes high while communicating, but may also be controlled by applications accessing the UART directly. To avoid problems with peripherals that require DTR (such as the

Husky Hunter 16 Printer Sidebox), you can set it to permanently on by setting AL to 1 and BL to non zero and then using this function.